

DATABASE_SCHEMA.md

EduMentor AI

Database Schema Design

Version: 2.0

Status: MVP

Overview

EduMentor AI uses:

- PostgreSQL
- Supabase
- pgvector

The database is responsible for:

- Authentication support
 - User profiles
 - Learning preferences
 - Learning sessions
 - Chat history
 - Quiz generation
 - Quiz results
 - Progress tracking
 - Analytics
 - API usage tracking
 - Future RAG support
-

Database Initialization

Enable pgvector before creating vector tables.

```
CREATE EXTENSION IF NOT EXISTS vector;
```

Entity Relationship Overview

```
profiles
|
|— user_preferences
|— conversations
|   |— messages
|— quizzes
|   |— quiz_results
|— progress
|— learning_analytics
|— api_usage

knowledge_chunks
```

Table: profiles

Stores user profile information.

```
CREATE TABLE profiles (
  id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,

  full_name TEXT NOT NULL,

  email TEXT UNIQUE NOT NULL,

  avatar_url TEXT,

  created_at TIMESTAMP DEFAULT NOW(),

  updated_at TIMESTAMP DEFAULT NOW()
);
```

Table: user_preferences

Stores onboarding responses.

```
CREATE TABLE user_preferences (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  user_id UUID REFERENCES profiles(id) ON DELETE CASCADE,  
  
  learning_topic TEXT NOT NULL,  
  
  skill_level TEXT NOT NULL,  
  
  learning_goal TEXT,  
  
  daily_study_minutes INTEGER,  
  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

Table: conversations

Stores learning sessions.

```
CREATE TABLE conversations (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  user_id UUID REFERENCES profiles(id) ON DELETE CASCADE,  
  
  title TEXT NOT NULL,  
  
  created_at TIMESTAMP DEFAULT NOW(),  
  
  updated_at TIMESTAMP DEFAULT NOW()  
);
```

Table: messages

Stores chat history.

```
CREATE TABLE messages (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
conversation_id UUID REFERENCES conversations(id) ON DELETE CASCADE,  
  
role TEXT NOT NULL CHECK (  
    role IN ('user', 'assistant')  
),  
  
content TEXT NOT NULL,  
  
created_at TIMESTAMP DEFAULT NOW()  
);
```

Table: quizzes

Stores generated quizzes.

```
CREATE TABLE quizzes (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    user_id UUID REFERENCES profiles(id) ON DELETE CASCADE,  
  
    topic TEXT NOT NULL,  
  
    difficulty TEXT NOT NULL,  
  
    quiz_data JSONB NOT NULL,  
  
    created_at TIMESTAMP DEFAULT NOW()  
);
```

Table: quiz_results

Stores completed quizzes.

```
CREATE TABLE quiz_results (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    quiz_id UUID REFERENCES quizzes(id) ON DELETE CASCADE,  
  
    user_id UUID REFERENCES profiles(id) ON DELETE CASCADE,
```

```
score INTEGER NOT NULL,  
  
total_questions INTEGER NOT NULL,  
  
percentage NUMERIC(5,2),  
  
submitted_at TIMESTAMP DEFAULT NOW()  
);
```

Table: progress

Tracks mastery by topic.

```
CREATE TABLE progress (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  user_id UUID REFERENCES profiles(id) ON DELETE CASCADE,  
  
  topic TEXT NOT NULL,  
  
  mastery_score NUMERIC(5,2) DEFAULT 0,  
  
  quizzes_completed INTEGER DEFAULT 0,  
  
  average_score NUMERIC(5,2) DEFAULT 0,  
  
  updated_at TIMESTAMP DEFAULT NOW()  
);
```

Table: learning_analytics

Stores dashboard metrics.

```
CREATE TABLE learning_analytics (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  user_id UUID REFERENCES profiles(id) ON DELETE CASCADE,  
  
  total_learning_sessions INTEGER DEFAULT 0,
```

```
total_messages INTEGER DEFAULT 0,  
  
total_quizzes INTEGER DEFAULT 0,  
  
average_quiz_score NUMERIC(5,2) DEFAULT 0,  
  
total_learning_minutes INTEGER DEFAULT 0,  
  
updated_at TIMESTAMP DEFAULT NOW()  
);
```

Table: api_usage

Used for rate limiting.

```
CREATE TABLE api_usage (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  user_id UUID REFERENCES profiles(id) ON DELETE CASCADE,  
  
  usage_date DATE NOT NULL,  
  
  chat_requests INTEGER DEFAULT 0,  
  
  quiz_generations INTEGER DEFAULT 0,  
  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

Table: knowledge_chunks

Future-ready RAG storage.

```
CREATE TABLE knowledge_chunks (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  topic TEXT NOT NULL,  
  
  title TEXT NOT NULL,
```

```
content TEXT NOT NULL,  
  
embedding VECTOR(384),  
  
created_at TIMESTAMP DEFAULT NOW()  
);
```

Adaptive Learning Logic

Quiz difficulty is determined using mastery score.

Mastery Score	Difficulty
0 - 40	Beginner
41 - 70	Intermediate
71 - 100	Advanced

Progress Calculation

Example formula:

```
Mastery Score =  
(  
Average Quiz Score × 0.7  
+  
Quiz Completion Consistency × 0.3  
)
```

Range:

```
0 - 100
```

Chat Context Strategy

To prevent context overflow:

Only send:

- Last 15 Messages
- Current User Message
- Relevant Knowledge Chunks (Future)

to the AI model.

Full history remains stored in PostgreSQL.

Database Indexes

Conversations

```
CREATE INDEX idx_conversations_user  
ON conversations(user_id);
```

Messages

```
CREATE INDEX idx_messages_conversation  
ON messages(conversation_id);
```

Progress

```
CREATE INDEX idx_progress_user  
ON progress(user_id);
```

Quizzes

```
CREATE INDEX idx_quizzes_user  
ON quizzes(user_id);
```

API Usage

```
CREATE INDEX idx_api_usage_user  
ON api_usage(user_id);
```


Vector Search

```
CREATE INDEX idx_knowledge_chunks_embedding
ON knowledge_chunks
USING ivfflat (embedding vector_cosine_ops);
```

Row Level Security (RLS)

RLS must be enabled on:

- profiles
- user_preferences
- conversations
- messages
- quizzes
- quiz_results
- progress
- learning_analytics
- api_usage

Example RLS Policy

```
CREATE POLICY "Users can access only their own data"
ON conversations
FOR ALL
USING (
    auth.uid() = user_id
);
```

Security Requirements

Users may only access:

- Their profile
- Their sessions
- Their messages
- Their quizzes
- Their analytics

Users must never access another user's data.

Final Database Summary

Table	Purpose
profiles	User Information
user_preferences	Onboarding Data
conversations	Learning Sessions
messages	Chat History
quizzes	Generated Quizzes
quiz_results	Quiz Results
progress	Mastery Tracking
learning_analytics	Dashboard Analytics
api_usage	Rate Limiting
knowledge_chunks	Future RAG Support

Total Tables: 10

Database Engine: PostgreSQL

Provider: Supabase

Vector Support: pgvector (384 dimensions)